DRUGS · CORTICOSTEROIDS

# Intratympanic steroids

Intratympanic steroids deliver dexamethasone or methylprednisolone directly to the middle ear, from where they diffuse through the round window into the inner ear, reaching perilymph concentrations far higher than oral treatment and with almost no systemic effect. They are used as salvage or primary treatment for sudden sensorineural hearing loss and, as a non-destructive option, for vertigo in Ménière's disease.

## 1 CHAPTER ONE At a glance

The treatment on one page.

### 1.1 Summary

INTRATYMPANIC STEROIDS

| | SUMMARY |
|---|---|
| **Agents** | Dexamethasone 10 mg/ml to 24 mg/ml; methylprednisolone 40 mg/ml to 62.5 mg/ml |
| **ENT uses** | Sudden sensorineural hearing loss: salvage after incomplete recovery with oral steroids, or primary treatment when systemic steroids are contraindicated (diabetes, psychiatric illness, intolerance); Ménière's disease (non-ablative vertigo control); autoimmune inner ear disease (adjunct) |
| **Regimen** | Commonly 0.4 to 0.8 ml injected under topical phenol anaesthesia, repeated 3 or 4 times over 2 weeks for sudden hearing loss; for Ménière's disease, methylprednisolone injections repeated as needed |
| **Timing** | Salvage within 2 to 6 weeks of onset of sudden hearing loss |
| **Risks** | Pain, transient vertigo, perforation (about 1% to 4%), otitis media |

# 2 CHAPTER TWO How it works

Mechanism and handling.

## 2.1 Pharmacokinetics

The round window membrane is the main route into the inner ear. Perilymph levels after intratympanic dexamethasone are many times those after oral or intravenous doses. Contact time matters: the patient lies with the ear up and avoids swallowing for about 20 to 30 minutes. A mucosal fold or a false round window membrane can block absorption.

# 3 CHAPTER THREE Using it in ENT

Indications.

## 3.1 Sudden sensorineural hearing loss

The AAO-HNS guideline recommends offering intratympanic steroid as **salvage** when recovery after initial treatment is incomplete, within 2 to 6 weeks of onset, and it is an option as primary treatment. Combined oral and intratympanic treatment is used in some centres for severe losses.

## 3.2 Ménière's disease

In a randomised trial, intratympanic methylprednisolone controlled vertigo about as well as intratympanic gentamicin, without the risk of hearing loss from the drug. It is therefore often tried before ablative gentamicin.

# 4 CHAPTER FOUR Safety

What goes wrong.

## 4.1 Adverse effects

Pain or burning during injection (less with dexamethasone than methylprednisolone), transient dizziness from cold or volume, **persistent perforation** in a small proportion (higher with repeated injections or a ventilation tube), otitis media, and dysgeusia from the chorda tympani.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The round window route and why the patient lies still.
- Indications in sudden hearing loss (salvage and primary) and in Ménière's disease.
- How the procedure is done and its risks.

## 5.2 Common pitfalls

COMMON PITFALLS

- Waiting more than 6 weeks before offering salvage.
- Forgetting to tell the patient not to swallow and to keep the head turned.

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